

Virtual Dream Reliving: Exploring Generative AI in Immersive Environment for Dream Re-experiencing

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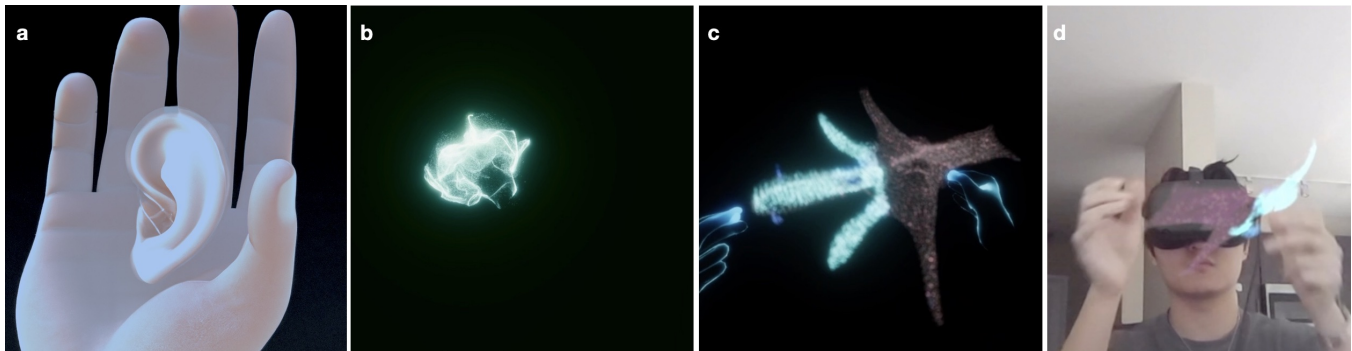


Figure 1: Overall Experience and Interaction: (a)The dreamer whispers one's dream into one's palm. (b) The words float into space as glowing particles. (c) The particles slowly form into a point cloud of the dream object. (d) The dreamer could use one's hands to directly manipulate the point cloud as a symbol for the dream object, and re-enact the story.

ABSTRACT

Dreaming is a fundamental component of the human experience. Modern-day psychologists and neuroscientists use “dreamwork” to describe a variety of strategies that deepen and engage with dreams. Re-experiencing the dream as if reliving the memory, feelings, and bodily sensations from the dream is a key element shared by many dreamwork practices. In this paper, we propose the concept of “dreamwork engineering” by creating a system enabling dream re-experiencing in a virtual reality environment through generative AI. Through an autoethnographic study, the first author documented his own dreams and relived his dream experiences

for two weeks. Based on our results, we propose a technology-aided dreamwork framework, where technology could potentially augment traditional dreamwork methods through *spatiality and movement, interactivity* and *abstract anchor*. We further highlight the collaborative role of technology in dreamwork and advocate that the scientific community could also benefit from dreaming and dreamwork for scientific creativity.

CCS CONCEPTS

• **Human-centered computing** → *Interaction design*; **Virtual reality**; • **Applied computing** → **Health care information systems**; • **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

Generative AI, Virtual Reality, Dreamwork Engineering, Dream Re-experiencing, Personal Insight, Scientific Creativity

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1 INTRODUCTION

Dreaming is a fundamental component of the human experience. Modern-day psychologists and neuroscientists use “dreamwork” to describe a variety of strategies that deepen and engage with dreams [14]. Beyond reducing depression [8] and increasing existential well-being [5], dreamwork could help dreamers attain personal insight [14]. Within the field of dreamwork, there has been a dramatic shift away from interpretation made by the dreamworker (or therapist), and towards more experiential methods enabling the dreamer to feel and re-engage with the dream experience. **Re-experiencing** the dream as if reliving the memory, feelings, and bodily sensations from the dream is a key contributing factor shared by many dreamwork practices including gestalt, Jungian, focusing, and Integral Dreaming [7].

Given its capacity for immersive multisensory storytelling and invoking profound bodily and emotional experiences, Virtual Reality (VR) has the potential to assist experiential dreamwork. VR has been used in previous dream research, such as for lucid dream training [24], incubating flying dreams [30] and simulating the feelings and benefits of transcendent flying dreams [27]. With the recent development of generative AI, especially 3D model generation, it is now possible to generate virtual objects and immersive environments from text-based dream records in near real-time. As a result, we were interested in how we might support dream re-experiencing through an AI-powered VR system. We present a novel exploratory VR system that synthesizes point cloud data from dream recordings with a diffusion AI model to allow immersive dream re-experiencing. The first author conducted a self-observation study and used the VR system to relive his dream experiences for 2 weeks. Based on our results, we propose a technology-aided dreamwork framework, and highlight the collaborative role of technology in dreamwork. We further advocate that the scientific community could also benefit from dreaming and dreamwork for scientific creativity.

2 RELATED WORK

2.1 Experiential Approaches of Dreamwork

Experiential re-entry into the dream is encouraged in almost all of the current occidental dreamwork methods [7]. The Hill Cognitive-Experiential method, a well-established model for dreamwork, involves three stages: Exploration, Insight, and Action [14]. In the exploration stage, therapists encourage clients to retell the dream, provide rich descriptions of dream images, associate them with dream images, and determine the triggers for the dream in waking life. In the insight stage, therapists help clients understand the meaning of the dream on at least one of four levels: the phenomenological experience of the dream, waking life, memories, or parts of self. In the action stage, the therapist helps the client talk about how she or he would like to change the dream. The therapist then

bridges the changes in the dream to changes in waking life. According to this model, the Exploration Stage is crucial in that it sets the foundation for the remainder of the process. Hill suggested that: “To arrive at a richer, more vivid dream interpretation, it may be valuable to stay within the dream and experience more details of the dream images, as well as to explore the thoughts, feelings, and memories that are associated with the dream images” [15]. She further suggested that immersing oneself in re-experiencing the dream might contribute to greater gains in exploration and insights.

We propose an experiential mechanism of dreamwork based on prior literature (see Fig.2). Experiential approaches include retelling the dream in the present tense, interacting creatively with the dream material (writing, drawing, enacting), and dream re-experiencing [7]. Ellis further concluded three practices for dream re-experiencing: finding a safe and relaxing dream object as “help” in a dream (from focusing), re-entering a dream element while taking on its perspective (from Gestalt), and enacting the continuation of a dream (from Jung) [6]. These approaches contribute to the experiential gain [13], along with dreamwork involvement, image exploration thoroughness and bodily sensations. The experiential gain then sets the foundation for insight gain and action gain.

As a result, this proposed mechanism motivated us to design a system that supports ultimate insight and action by first enabling dreamers to engage and re-experience one’s own dreams.

2.2 Dream Engineering and Virtual Reality

Dream engineering [2] refers to the use of technologies to monitor and attempt to influence people’s dream experiences for memory enhancement, creativity, emotion regulation and physical rehabilitation, especially using sensory stimulation. Within dream engineering, HCI communities have explored dream incubation [12], lucid dream training [24], dream tracking and dream informatics [17], and recreating the dream-flying experience in VR [27]. Most of these dream engineering works focus on the stages before the dream (priming, training), or during the dream (dream incubation), but few have explored how technology might be used *after* the dream, supporting the process by which dreamers can engage and re-experience one’s own dream to develop further personal insights. We thus propose a new subfield of “dreamwork engineering”, which uses technologies to support various approaches in dreamwork for experiential gain, insights, and actions.

In addition to dream engineering work using VR as a multi-sensory stimulation before sleep [24, 30], there seems to be some parallel between VR and dream theory which could inform the VR system design for dreamwork engineering. According to Windt’s immersive spatiotemporal hallucination model of dreaming (ISTH), dreams minimally involve a “here and now” feeling. One critical dimension of Windt’s oneiragogic spectrum is the creation of a visuospatial scene, which induces a sense of being located at a specific point in a larger spatial expanse [36]. Traditional dreamwork methods are based on the dreamer’s ability to re-imagine the dream scenario through visualization, reading a dream journal, listening to a dream recording, or retelling the dream in the present tense in detail, which has some natural limitations on re-experiencing dream phenomenon spatially and immersively. The spatiality offered by VR systems could thus potentially contribute

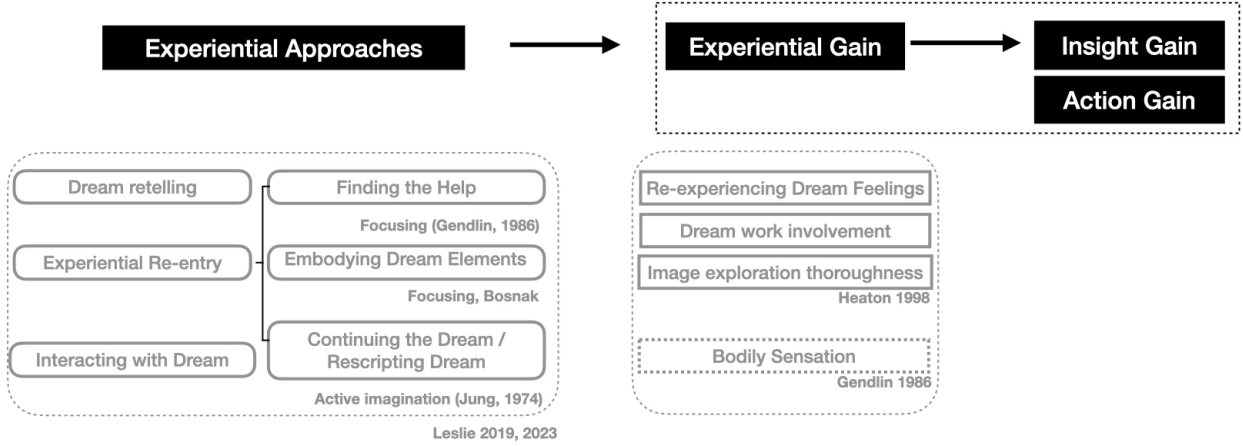


Figure 2: Proposed Mechanism of experiential dreamwork for insights and actions.

to the dream re-experiencing by resembling the spatial quality in dreaming phenomenon and immersing the dreamer into a full "here-and-now" experience. This motivated us to use VR as a technology to re-experience the visuospatial aspect of dreams.

2.3 Generative AI for 3D Scene Generation in VR

Anticipating the emergence of language-based 3D scene generation systems, Coyne and Sproat [4] presented WordsEye, a tool that automatically converts text into 3D scenes. Along with this method, more recent work has explored natural language commands for scene generation [3]. However, these tools relied on a vast database of preexisting 3D models and poses. In contrast, fictional world-building typically involves the creation of new artwork, where this method may face limitations. As a result, integrating 3D generative AI could serve as a game changer to break this limitation of prefabricating 3D models.

The field of generative AI focuses on developing algorithms and models capable of generating synthetic data that closely resemble real-world data, including text, 2D images, and even 3D content [11]. At the intersection between VR and 3D Generative AI, prior research includes helping designers to explore 3D Generative Design solutions with VR as a computational design tool [22, 23, 35], and helping VR developers to author VR content and interactions while immersed in virtual environment [38].

In terms of VR scene generation through AI, the closest work to our knowledge is the GetWild system [37]. In their system, the user can generate 3D objects by capturing photos of real-life objects. The user can then use VR to modify those AI-generated 3D Objects, by editing object vertices, pasting materials and adding particle effects. This system also allows the user to use AI to generate a 3D terrain based on one's 2D sketch. However, the system is far from real-time generation, and the image to 3D AI model is bounded by the reference photos taken from the real world.

To overcome these limitations, here we chose a text-to-3D generative AI model, *Point-E* by Open AI, considering the trade-off between generating time and model quality [28]. The single-run

diffusion model *Point-E* model used in our system generates a point cloud with 4096 resolution in 17 seconds, while the multiple-run diffusion models such as *dreamfield* model takes around 1 hour to generate a higher quality model [20]. On the one hand, we chose the *Point-E* model over other generative models because we envisioned a near real-time interactive system to walk through dream objects. On the other hand, dream objects are usually ambiguous and, as designers, we could take advantage of the aesthetic from lower-resolution point cloud models to lean into the design constraints. That is, instead of pursuing a photorealistic rendering of the dream object, we could explore an abstract aesthetic to allow space for the user to attribute meaning.

In addition, to allow for a more natural interaction with the text-driven 3D generative AI, we decided to use speech recognition in VR, triggered by a gesture of palms facing up.

As a result, our system contributes to the discourse by allowing users to create AI-generated 3D objects in VR through speech, without creating a database of preexisting 3D models. Moreover, our system allows for near real-time VR content generation and opens up spaces for further user interactions in VR.

3 METHOD

3.1 Technical Apparatus

The VR experience was created using the Unity game engine. First, a speech recognition function provided by Meta voice SDK transcribes the dream object description in Unity, which is then sent to the server. The server relays the text description as prompt input for the *Point-E* model [28]. The generated 3D model is then uploaded to the server in PLY format. This file is fetched from the server into Unity, with the position map and color map data parsed and used for the point cloud rendering in Unity Engine (see Fig.3). The program was run on a portable desktop computer and streamed to an Oculus Quest HMD with Oculus Link.

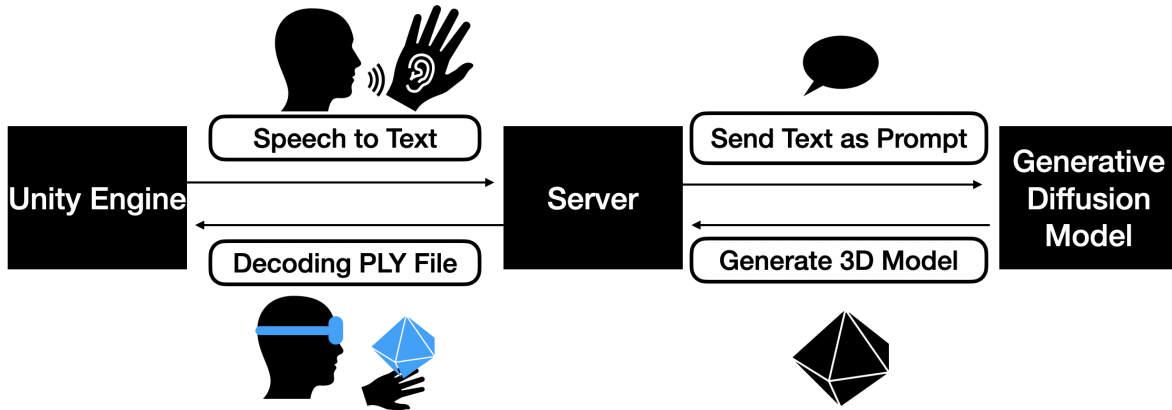


Figure 3: System Setup

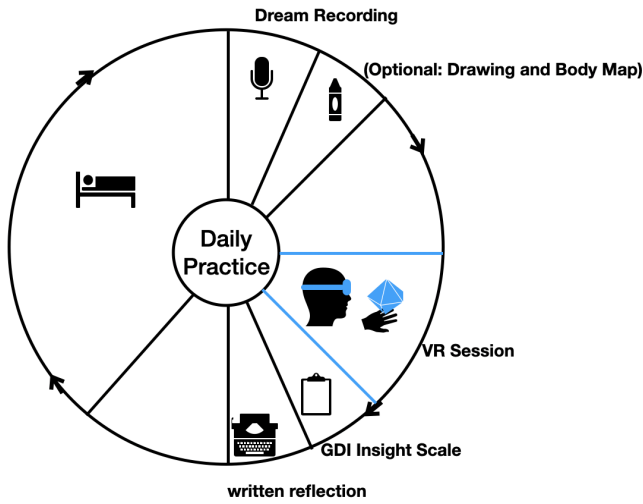


Figure 4: The protocol for self-observation and documentation of the daily practice.

3.2 Self Observation Protocol: Dream Documentation and Reliving

The HCI community has a long tradition of almost exclusively producing “objective ethnographies”, attempting to generate accurate descriptions of the “world” and the individuals inhabiting it [31]. However, we have observed a movement towards recognizing the value of the subjective lived experience. Autoethnography is one first-person method in which the fieldworker’s experience is investigated together with the experience of other observed social actors. The researcher’s personal account is considered valuable on its own and worthy of reporting in the ethnographic recounting [34]. The first-person approach could provide unique data and insights otherwise unavailable, especially in personal, domestic, or intimate contexts (such as dreaming). Goodall [10] emphasizes how autoethnography “completely dissolves any idea of distance, doesn’t produce ‘findings,’ isn’t generalizable, and only has credibility when self-reflexive, and authority when richly vulnerable...

When it is done well, we can learn previously unspoken, unknown things about culture and communication from it”.

In this paper, an autoethnographic study was carried out by the first author, a researcher and media artist interested in dreams and altered states of consciousness since 2019. He has been recording his own dreams sporadically on a monthly basis when he came across a profound dream, and has experimented with dream reimmersion through listening to dream recordings and reading dream journals. He had never recorded his dreams on a daily basis before the study.

During the study, the first author lived with the system for two weeks. Each morning, he turned on a voice recording app and recorded his own dreams if he could remember any. In addition, he did optional dream drawing or body mapping [19] if he came across vivid imagery or strong bodily sensations. Over 2 weeks, he recalled and recorded dreams on 4 nights. From these 4 nights, he had 10 dream images with distinct scenes ($N = 10$). At the end of each day, the first author used the VR system to relive his dream experiences, and video recorded his behavior and the on-screen VR

experience. He then completed Heaton's Gains from Dream Interpretation (GDI) questionnaire [13] and used each item as a prompt for his written reflection. The GDI scale has 14 items using a 9-point scale (1–9, with descriptors "strongly disagree" to "strongly agree"). This questionnaire comprises 3 subscales: Exploration-Insight gains, Experiential gains, and Action gains.

To measure the quality of dream re-experiencing, we used the Experiential gain subscale (comprising items 3 and 9), along with items 1 and 8 from the Exploration subscale on dream exploration thoroughness and dreamwork involvement:

3. During the session I was able to **re-experience the feelings** I had in my dream.

9. I felt like I was **actually reliving** the dream during the session.

1. I was able to explore my dream **thoroughly** during the session.

8. I felt like I was very **involved** in working with the dream during the session.

To assess personal insight more specifically, we used items from the Insight subscale (2, 6, 7, 12, 13) along with one item from the Action subscale in which an insight leads to a change in a way of thinking (10):

2. I learned more about what this dream meant for me personally during the session

6. I learned more from the session about how past events influence my present behavior.

7. I learned more about issues in my waking life from working with the dream.

10. I learned a new way of thinking about myself and my problems.

12. I learned things that I would not have thought of on my own.

13. I was able to make some connections between images in my dream and issues in my waking life that I had not previously considered.

We used items from the Action gains subscale other than item 10 to measure action gain:

4. I have more of a sense that I can change my dream when they are frightening or bad.

5. I got ideas during the session for how to change some aspect(s) of myself or my life.

11. I will use the things that I learned in this dream interpretation in my life.

14. I felt reassured about myself or my dream after this session."

In total, the first author did VR-assisted dreamwork on 10 dream images. The first part of the Results section will describe the general experience of the dreamer when generating dream objects in VR. The second part of the Results section will describe in more depth the whole dreamwork process of one selected dream. In fact, many dreamwork practices encourage the dreamer to engage with the images that evoke the strongest emotion, and sometimes with the single most significant image [7]. Thus, the first author selected the 3 most significant dream images, all coming from the same night (Night 4), as a case study in this paper for dream re-experiencing. As the first author is the dreamer engaging in the daily documentation and reflection, the pronoun 'I' and present tense will be used to speak about the personal experience of the practice in the following sections.

4 RESULTS

4.1 General Experience

During the self-observation study, the first author generated 46 dream objects with prompts for the 10 dream images, which included characters (e.g., "murderer", "police"), objects (e.g., "a Khaki sofa bed"), settings (e.g., "Tunnel", "The big boat"), and concepts (e.g., "Escape") (see Fig.5). For each dream image or dream scene, the first author prompted at least 2 and at most 10 dream objects ($M = 4.60, SD = 3.10$). For each object, the first author often used one or two simple words for description ($M = 2.20, SD = 1.13, Min = 1, Max = 4$).

When generating dream objects, simple adjectives about color and size with a concrete everyday object usually yielded more satisfactory results (e.g., "a blue bus" or "a high chair"), but less satisfactory when the descriptive includes plural quantifier ("a row of seats") and more abstract two-dimensional objects ("fingerprint", "navigation map"). When generating simple settings, the model was also usually satisfactory (e.g., "stadium" or "boat"), and the user could then scale up the model to get inside the abstract point cloud. When generating concepts, not surprisingly the results were often obscure and seemed unrelated to the semantic meaning of the prompt (e.g., "Escape", "Game Design" or "Diagram"). When generating human characters, it seemed easier when prompting adjectives with clear visual features such as "police" (uniform) or "blonde" (color), but it seemed difficult to generate characters with vague visual features such as "murderer" or "mother". In summary, it seemed easier for the user to use simple descriptions about size and color to generate concrete everyday objects and generic human characters, but less so for abstract concepts, two-dimensional objects and plural quantifiers.

4.2 The Dream Reliving Case Study

4.2.1 Dream Journal.

Dream Image 1. I am running to catch the buses in front of a hospital, one in red and one in blue, but both of them drive away. I take out my phone to look up the map, and decide to sit down and wait. As I sit back, I find myself sitting in the backseat of a taxi. A mother and her boy get in the taxi, and the driver starts the engine. I get out of the taxi hurriedly, yet all my keys and wallet are left on the seat and I have to recollect them one by one. The people in the taxi tell me I could contact them anytime if I miss anything. So I take out my phone and try to take a picture of the license plate for record. Three number '3' shows on my lock screen instead of my everyday lock screen picture (see Fig.6(a)). The thought of 'I might be dreaming' flashes by and slips away swiftly from my dream ego consciousness. I try to look closer at the license plate number, but the number keeps changing and elongating.

Dream Image 2. Nothing in sight, just white light. I hear dishes clinking, and my family's voice in the background. Sounds like I'm at the dining table at home with my family. Then my mom's voice casually mentions that my grandma is in the shopping mall. I am shocked because I know in reality my grandma passed away 5 years ago. So I ask the voice: "So grandma hasn't left (this world) yet?" And then my mom's voice comes from a distance: "No no she has." I am confused and I realize that maybe Grandma is still here in the

N1	N2	N3	N4	
Indian pancake Cupboard 1 Amazon website 2 Computer monitor The big boat Game design 3	Murder Tunnel Rooftop Escape Police 4 Murderer A high chair A khaki sofa bed 5	Dan Stadium A row of seats Ear 6 A blonde girl Strange city In bullets Army's Fingers Fingerprint A rail of seats Benches Mobile phone 7	A two floor hospital A red bust A blue bus My mobile phone Navigation map A taxi There's a mother And a little boy The female taxi driver 8 The blue license plate A dining table 9 With the kitchen things Information diagram A canvas board Diagram A wooden board Sculptures Canvas Crystal ball 10	Object Character Setting Concept

Figure 5: An overview of Prompts used in dream reliving in the self-observation study. Each block is one night of dreams, ranging from N1 to N4. Each number with the sub-block indicates the dream image index. The prompts are color-coded as object(black), character(purple, setting(blue) and concept(green).



Figure 6: Hand drawings of some visual dream imageries.

dream world. And maybe, if I stay here long enough and keep the conversation going, I can get to talk to Grandma. Meanwhile, I feel my consciousness slowly withdraw from the dream. The dream is collapsing. Just hang on for a bit longer, I tell myself. I feel that my body sink back down, like I'm diving back into the water of the dream. As the water top over my head, I feel a humming sound in my head, and a numbness or thickening on my scalp (see 7(a)). The vision comes back to the white light and my mother's voice becomes clear again. As my consciousness floats up and down in this water of dreams, I start to hallucinate and hear my grandma's voice from my childhood memory. This voice of memory seems to come over from an old telephone, and it keeps calling my nickname "Tong Tong". A voice full of love, again and again. And every time I hear it I feel my eyes burning, my chest tight with something blocked inside, and a boiling warmth rising up from my heart to my eyes (see Fig.7(b)). I'm on the edge of breaking into tears. Meanwhile I try to keep asking my mom more about grandma, but I start to lose my voice till I can't make any sound.

Dream Image 3. I wake up from the dream and I am in a meeting with my co-author, discussing this dream. I tell him I wish I could talk to my grandma, because I know many people tried to talk to

dream characters, such as god or family members who passed away, and this might shed some light on this ongoing dream-reliving project. Then I came up with a visual analysis of the dream, a digitally animated diagram (Fig.6(b)). The diagram seems to evolve intelligently in real-time based on the dream data. Maybe I can use it in my paper, I thought to myself. So I decided to take a picture of the diagram. The diagram is attached to a wooden canvas, and as I try to turn the canvas to get a better angle, I find it surprisingly heavy. It turns out there's a hidden layer of canvas beneath the diagram (Fig.6(d)). I can see 2 by 3 docks on the second layer, and some mysterious physical objects placed in each dock – e.g., wooden sculptures, or translucent crystal balls. Now again the dream is collapsing, I realize the environment is a European-style museum or library (Fig.6(c)), before everything is pulled away from me.

4.2.2 Re-entering the Dreamscape. I sit at my desk in my living room and put on my VR headset. With the headset on, I start to play back my own dream voice recording. I personally found that listening to my own voice from my dream recording could help me quickly transition into a hypnagogic state. After 5 minutes of listening to my recording, I start to retell my own dream in the present tense, first-person perspective. Whenever it comes to a



Figure 7: Body Map illustrating the bodily sensations in Dream Image 2.

concrete dream object, I put my palms up together, and a tiny ear grows in my palm. I then whisper the dream object into the tiny ear (see Fig.1(a)). The words I've spoken show up in the air. After 25 - 30 seconds, the dream object appears in the air (see Fig.1(b)). I can then manipulate the scale and position of the dream object using my hand or a controller (see Fig.1(c)(d)). Each object has physics with zero gravity applied and a collider attached. Since I found occasional loss of hand tracking in my earlier pilot test, in this self-observation study I used hand controllers for direct manipulation. In the following description, I use the word *summon* as a shortening for my action of whispering dream object description until its appearance. I use quotation marks to indicate the transcribed content as prompts. During the session, I disregarded generated 3D models from mistranscription and accidental gesture triggers.

For Image 1, I summon "a two-floor hospital", "a blue bus" (see Fig.8(a)), and a wrongly transcribed "red bust". I then rescale the hospital and the bus to mock up my perspective when I was in the dream (see Fig.8(b)). I then pick up "my mobile phone", and try to re-enter the position where I just missed the bus and try to look up the navigation (see Fig.8(c)). I then describe and move the "taxi" along with the "mother" with her "little boy" and "female taxi driver". I try to fit these figures into the taxi (grouping them together) but I fail because of the colliders attached to each model.

So I re-position the taxi and mock up the scene where I use my phone to take a picture of the license plate (see Fig.8(d)).

For Image 2, although I don't have any visual image in my dreams, it seems I subconsciously visualize the context by summoning "a dining table" and "the kitchen things" indicated by the clinking sound (see Fig.8(e)). Being that the visual is minimal, I can still feel the strong emotions and bodily sensations as I am narrating the dream memory within the VR experience.

For Image 3, I try to summon an "information diagram" but only get a shape I could not recognize. I then summon "a canvas board" as an abstract representation of the diagram, and a "wooden board" as the second hidden layer beneath the canvas. I then summon "sculptures" and "crystal balls" as the representation of the concepts of the physical objects attached to the second board. Then again I try to group these objects closer to the double-layer canvas I saw in my dream, but the collider and physics make the objects fly around in chaos and I have to give up in frustration.

5 DISCUSSION: A FRAMEWORK TOWARDS TECHNOLOGY-AIDED EXPERIENTIAL DREAMWORK

We discuss reflection on the effectiveness of the VR experience using the GDI insight scale as prompts, and considerations on what

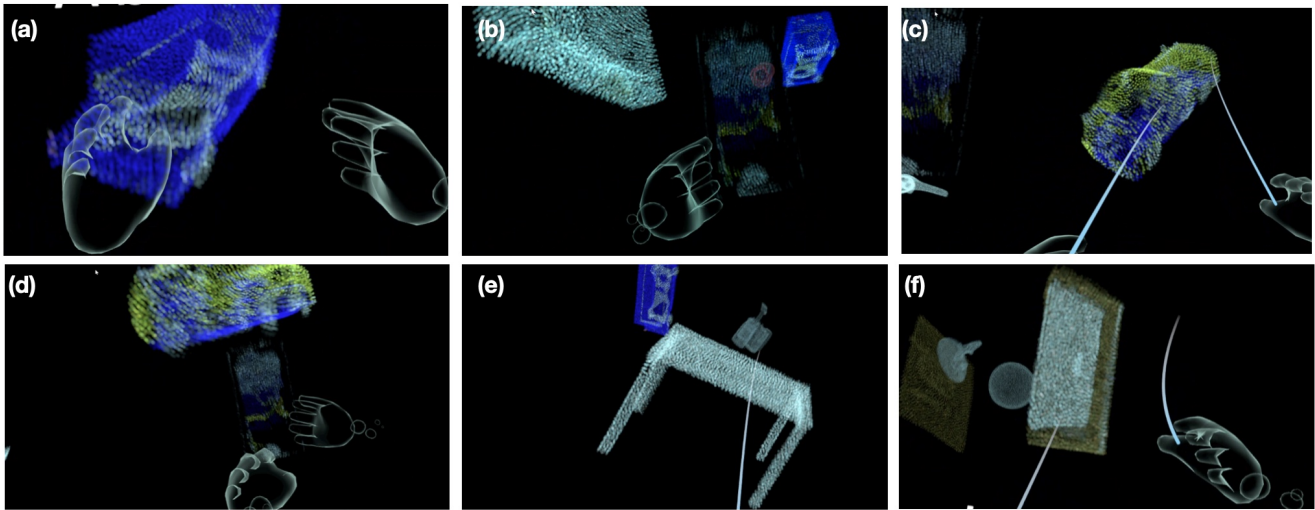


Figure 8: (a) I use the controller to rescale and re-position the "blue bus". (b) I take out my "mobile phone" to look up the map. (c) I use the raycasting to rescale and move the "taxi" around. (d) I try to take a picture of the taxi license plate from behind. (e) I visualize a family "dining table" with "kitchen things". (f) I try to group the double-layer canvas and the physical objects.

aspects of the AI-powered VR system could potentially contribute to experiential, insight, or action gain. We then build on the previous mechanism (Fig.2) and propose an experiential framework that combines Ellis's experiential lens of dreamwork and Hill's model, with an emphasis on technology's role (see Fig.9). We propose that the technological system could augment the experiential approaches of dreamwork proposed by Ellis [6], and further contribute to the experiential gain, through *spatiality and movement, interactivity and abstract anchor*. Meanwhile, we reflect on the limitations of the model and space for improvement. In addition, we propose that the generative AI-powered system collaborates with the dreamer to facilitate insight gain through fully re-experiencing the dream feelings. Finally, the scientific community might also benefit from dreaming and dreamwork for scientific creativity.

5.1 Augmenting Experiential Approaches through Technology: Potential Contributors

5.1.1 Spatiality and Movement Might Contribute to Dream Re-experiencing. In Dream Image 1, I reported high ratings on experiential gain (6.5), image exploration thoroughness (7), dreamwork involvement (8), and self-reassurance (7). In reflection, I gave high ratings on thoroughness and involvement because in the VR experience I was able to mock up my visual perspective that is similar to when I was inside the dream (Fig.8(b)), which also reflected our design intent of leveraging the **spatiality** in VR.

In addition, during the VR session, bodily actions of holding a prop (e.g., mobile phone as in Fig.8(d)), as well as moving and scaling the dream objects seemed to contribute to a sense of agency and self-reassurance that was related to the experiential gain. This finding is aligned with our previous work on simulating a flying dream experience in VR, which suggests that embodied interaction in VR contributes to feelings of dream-flying through movement efficacy, competent feelings due to the dreamer's exceptional ability

of movement in dreams [27]. This result is also in parallel with the notion that kinesthetic imagery is a primary aspect of dream experiences[29], which can metaphorically represent feelings and emotions in an embodied way. We speculate that **movement** in dreams might be a key contributing element to the experiential aspects of dream feelings and emotions. Correspondingly, in VR sessions, the spatiality and movements might be one key element to experiential gain by augmenting the experiential approaches "Dream Retelling" and "Experiential Re-entry".

5.1.2 A System Allowing for Interactivity with Dream elements. In Image 1, the direct manipulation of the dream objects (grabbing, moving, scaling) allowed me to **interact** with the dream elements and remap their relationships. In dreamwork, there has been a trend of treating the dream not as fixed content but rather as alive, fluid, and interactive [7]. Many dreamwork approaches have encouraged the dreamer to dream the dream onward, which refers to any way that the dream has evolved so that it is no longer the same dream [6]. In Hill's model, the dreamworker guides the dreamer to discuss how one would like to change the dream, and then bridges from the changes in the dream to changes in waking life [14]. In Imagery Rehearsal Therapy (IRT) [25], dreamers are asked to change their nightmare in any way they wish and to rehearse this new version of the dream during the day and before sleep. Similarly in the VR session, by actively changing the position, scale and spatial relationship between the dream objects in VR, the dreamer could potentially keep reshaping the dream space and let the dream evolve, compared to passively watching a fixed representation of the dream representation.

This interactivity afforded by our system potentially augments the experiential approaches "Interacting with Dream" and "continuing the dream", which might be another contributing element towards experiential dreamwork.

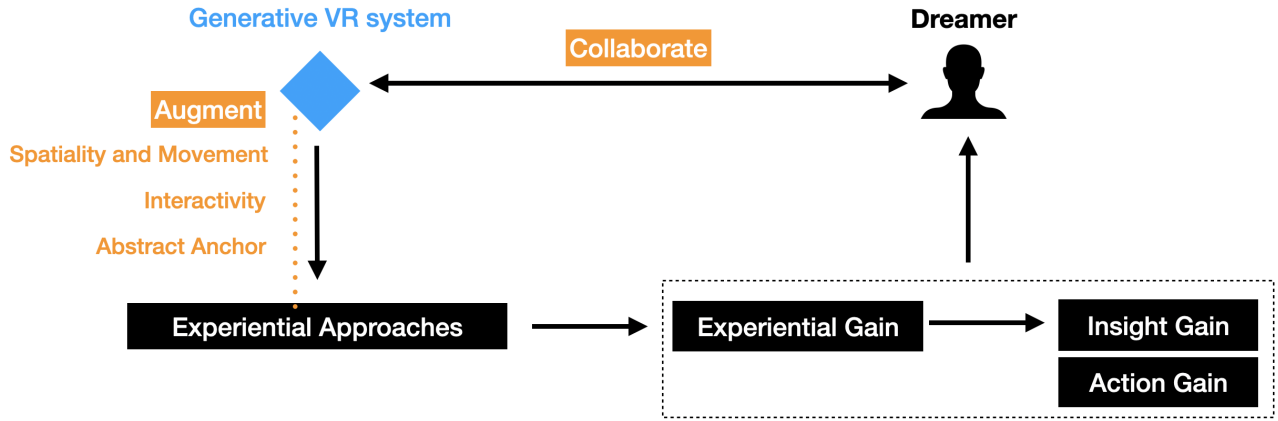


Figure 9: Technology-aided experiential dreamwork framework.

5.1.3 Abstracting AI-Generated Object to anchor the concept of a personal dream object. In Image 2, I similarly reported high ratings on experiential gain (7), image exploration thoroughness (7), dreamwork involvement (8), and self-reassurance (7). Although there was minimal summoning and interactivity in exploring this image, I reported the high ratings because I could feel strong bodily sensations and emotions that were akin to what I experienced in my dream.

In VR, I visualized the family dining table even though I had no visual imagery during the dream. In reflection, this could be potentially because the morphing and translucent AI-generated object served as an abstract yet personalized **anchor**, which I could then attribute a personal meaning of family and a sense of safety.

This abstract anchor then *"helped"* me set up the mindset and provided me with a leap-off place for re-experiencing strong emotions. As a result, the system in a way **augmented** the experiential strategy of *"finding the help"*, and potentially contributed further to the experiential gain.

To summarize, our system seemed to augment dream re-experiencing through spatiality and movement, interactivity and abstract anchors, which potentially further contributed to the experiential gain in dreamwork. For future work, researchers could consider further augmenting other practices for dream re-experiencing, such as *"embodying the dream element"* and *"continuing the dream"*.

5.1.4 Tension Between Prompt Engineering and Natural Dream Narration. One limitation of the system is that many descriptions of emotions and concepts from dream reports could not easily be visualized as 3D objects in VR. While these pieces of information were usually crucial to the dream experience, they were missing in the VR experience. Moreover, in his reflection notes the first author noticed a pattern of self-censoring when using the system by always using short transcription and censoring parts of the dream experience in order to provide simpler and clearer prompts for the diffusion model. This awareness of the model's limitation and self-censoring processes thus created a tension between naturally narrating one's dream in detail and prompting with clear instructions.

In future work, researchers could consider using Large Language Models to translate natural narration into clear and structured

prompts for the diffusion model. A follow-up study with a larger and more diverse sample size could help us evaluate whether or how these identified contributors affect dream re-experiencing, and understand how the mistranscriptions and the missing objects, crucial for the dream experience, affect the reliving of the dream. While the current prototype focuses on the visuospatial aspect, future iterations could explore how other sensory elements (auditory, tactile, temperature, olfactory) impact dream re-experiencing. The dream research community also needs to attend to ethical considerations of recreating deeply personal experiences, and instrumentalizing the private dream space of rest and reflection.

5.2 Generative AI as a collaborator for dreamers to derive autonomous insights

Within dreamwork, there has been a paradigm shift from interpretative towards collaborative – the dreamer as an expert in determining one's own dreams, and the dreamworker as a collaborator to help dreamers reveal the locked answer within themselves [7]. There is a strong agreement that it is better for dreamers to discover a dream's meaning through dream image exploration, rather than having an interpretation "fed" to them by a dreamworker, because "then the meaning has personal salience and is something that they can integrate into their lives" [16]. Thus the generative AI-powered system might serve as another **collaborator**, who facilitates the dreamer to derive insights independently through fully re-experiencing the dream feelings. However, the intent is not for our system to fully replace the role of a dreamworker in this process. Within our framework, we can imagine a scenario where the dreamworker is also present, who then collaborates with the AI-powered VR system and together helps the dreamer explore the meaning of the dream.

5.3 Scientific Creativity from Dreams

In the case study, Dream Image 3 involves my own research, and my interpretation leads to a new understanding of the research-creation and even initiated a new design in my research-creation. This resulted in higher ratings of Action gain (6) compared with Image 1 (4.25) and Image 2 (3.75), and a higher personal insight gain (4.83) compared with Image 1 (3) and Image2 (2.5). I interpreted the

image as my inner struggle between digital media and physicality within my research. As an artist-researcher working with VR as a digital medium, I have been exchanging ideas intensely with my colleagues from a sculpturing and installation background in the past two months. These exchanges have helped me shift more attention towards physicality. I see the digital canvas in my dream as a metaphor for the digital medium I have been working with, and the double-layer canvas reveals that there is a hidden layer of physicality and materiality behind the digital medium. Those physical sculptures seem to be a physical representation of the dream objects and ask me to think beyond the digital presentation of my research-creation. This new understanding then pushes me to design a dream pod as an installation for the VR experience and helps me steer the research-creation toward a more physical direction.

This personal experience of scientific insight when using the system is in line with previous work showing that sleep and dreaming can bolster problem-solving and creativity [1, 18, 26, 33]. Modern science has often underestimated and neglected adventurous and associative modes of thinking with an overemphasis on analytical thinking [32]. However, scientific innovation, invention and discovery can also arise from unconstructed thinking, dreams and flashes of insight. One example is from the 19th-century German chemist Friedrich August Kekulé, who had a hypnagogic reverie where he saw a snake seizing its own tail, which inspired him to discover the ring structure of benzene, and hence the structure of all aromatic compounds. Other scientists, such as Albert Einstein, Neils Bohr, Archimedes and Dmitri Mendeleev, claimed similar creative benefits from their daydreaming minds. Beyond dreaming's natural hyperassociative state, recent research has also proposed psychedelics as another potential catalyst of scientific creativity and insight, and are starting to better understand the underlying neurophenomenological and cognitive processes involved [9]. We feel it is important that we HCI researchers better understand how dreams and other altered-states of consciousness might support scientific creativity and we advocate for more discussions on this topic in the field. Chemist August Kekulé once said at a scientific conference, reflecting on his seminal hypnagogic insight: "Let us learn to dream, gentlemen, and then perhaps we shall learn the truth ... but let us beware of publishing our dreams before they have been put to the proof by the waking understanding" [21].

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